



# Polysaccharides from *Ganoderma lucidum* attenuate cognitive impairment in 5x FAD mice by inhibiting oxidative stress and modulating mitochondrial dynamics via the Nrf2/antioxidative axis activation

Xiaoqin Liu<sup>1</sup> · Yanbing Li<sup>2</sup> · Jiwei Wang<sup>2</sup> · Tao Meng<sup>1</sup> · Lijuan Song<sup>2</sup> · Lizhi Yang<sup>3</sup> · Jiezhong Yu<sup>1,2,4</sup> · Cungen Ma<sup>2</sup>

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## Abstract

Oxidative stress and mitochondrial dynamics imbalance are key contributors to AD pathogenesis. GLPS, an extract from *Ganoderma lucidum* spores, exhibits anti-inflammatory, antioxidant, and immunomodulatory properties. However, the roles of GLPS in regulating oxidative stress and mitochondrial dynamics in AD remain poorly understood. Here, the underlying mechanisms of neuroprotective effects on cognitive dysfunction in 5 × FAD mice were explored. C57BL/6 mice served as WT controls, while 5 × FAD mice were divided into an AD group and an AD + GLPS group. The mice in AD + GLPS group were administered daily GLPS (25 mg/kg) by i.p. injection for two months, while WT and AD mice received an equivalent volume of normal saline. The results indicated that GLPS markedly improved cognitive function and decreased p-tau and Aβ levels in 5 × FAD mice. Moreover, GLPS alleviated oxidative stress by increasing SOD levels and decreasing MDA concentrations. It also inhibited excessive mitochondrial fragmentation by decreasing the expression of p-Drp1 and Fis1, while increasing the levels of Mfn1, Mfn2, and OPA1 in 5 × FAD mice. Mechanistically, GLPS activated Nrf2, leading to a marked upregulation of antioxidant enzymes, including HO-1, NQO1, and SOD2 in 5 × FAD mice. Collectively, these findings suggest that GLPS ameliorates cognitive deficits in 5 × FAD mice by reducing oxidative stress and modulating mitochondrial dynamics through Nrf2-mediated antioxidant enzyme activation.

**Keywords** GLPS · Oxidative stress · Mitochondrial dynamics · Nrf2/antioxidative pathway · AD

Xiaoqin Liu and Yanbing Li contributed equally to this work.

✉ Jiezhong Yu  
sxdtyjz@qq.com

✉ Cungen Ma  
macungen@sxtcm.edu.cn

<sup>1</sup> Institute of Brain Science, Medical School, Shanxi Datong University, Datong 037009, China

<sup>2</sup> The Key Research Laboratory of Benefiting Qi for Acting Blood Circulation Method to Treat Multiple Sclerosis of State Administration of Traditional Chinese Medicine, Research Center of Neurobiology, Shanxi University of Chinese Medicine, Jinzhong 030619, China

<sup>3</sup> Shanxi Guorun Pharmaceutical Co., Ltd, Datong 038100, China

<sup>4</sup> Department of Neurology, The Fifth People's Hospital, Datong 037009, China

## Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the most common cause of dementia in the elderly. Its hallmark pathological features include the accumulation of extracellular beta-amyloid (Aβ) plaques and intracellular hyperphosphorylated tau (p-tau) neurofibrillary tangles, both of which contribute to the synaptic loss, neuronal death, and cognitive decline (Tzioras et al. 2023). Aging is the primary risk factor for AD, and with the global rise in the elderly population, the prevalence of AD is expected to increase, exacerbating its burden on public health and the economy. Despite extensive research, AD remains incurable, necessitating further investigation into its underlying mechanisms and potential therapeutic strategies.

Emerging evidence suggests that oxidative stress and mitochondrial dynamics dysregulation play crucial roles in AD progression (Nabi et al. 2022). Oxidative stress results

from excessive reactive oxygen species (ROS) production, primarily by mitochondria, leading to oxidative damage and mitochondrial fragmentation (Kang 2020). Studies have reported altered oxidative stress biomarkers in AD, including elevated malondialdehyde (MDA) levels in brain tissues and significantly reduced superoxide dismutase (SOD) activity in the superior temporal gyrus of patients with AD compared to controls (Youssef et al. 2018; Hannan et al. 2020). Mitochondria are central to cellular energy metabolism, and excessive mitochondrial fragmentation contributes to ROS overproduction (Jezek et al. 2018; Casanova et al. 2023). Mitochondrial morphology and function are regulated by dynamic fission and fusion processes. Dynamin-related protein 1 (Drp1) is the key regulator of mitochondrial fission and is activated by phosphorylation at serine 616. Other adaptor proteins, such as fission 1 (Fis1) and mitochondrial fission factor (Mff), also facilitate fission. Conversely, mitochondrial fusion is mediated by optic atrophy 1 (Opa1), mitofusin 1 (Mfn1), and mitofusin 2 (Mfn2) (Liu et al. 2022; Zhang et al. 2023). Notably, AD brains exhibit increased Drp1 phosphorylation at serine 616 and elevated Fis1 expression, alongside significantly reduced Mfn1, Mfn2, and OPA1 levels (Wang et al. 2009; Cenini and Voos 2019). Inhibition of Drp1 using Mdivi-1 has been shown to mitigate mitochondrial fragmentation, improve cognitive function, and reduce A $\beta$  deposition and oxidative stress in AD mouse models (Wang et al. 2017). Oxidative stress and alteration of mitochondrial dynamic are mutually intertwined pathological processes, thereby promoting the pathology of AD. Therefore, identifying novel therapeutic agents targeting these pathological processes could help unravel neuroprotective benefits and advance AD treatment.

Growing evidence suggests that nuclear factor erythroid 2-related factor 2 (Nrf2) plays a critical cytoprotective role in mitigating neurodegeneration by regulating oxidative stress and mitochondrial dynamics (Ammal Kaidery et al. 2019; Villavicencio Tejo and Quintanilla 2021). As a transcription factor, Nrf2 induces the expression of antioxidant enzymes, including heme oxygenase 1 (HO-1), NAD(P)H quinone dehydrogenase 1 (NQO1), and superoxide dismutase 2 (SOD2) (Zhang et al. 2021). However, both aging and excessive ROS reduce Nrf2 activity, impairing the antioxidant defense system (Villavicencio Tejo and Quintanilla 2021; Zhao et al. 2023a, b). Studies have reported decreased Nrf2 levels in patients with AD and AD mouse models, with Nrf2 ablation exacerbating cognitive deficits and worsening AD pathology (Rojo et al. 2017; Villavicencio Tejo and Quintanilla 2021; Jiang et al. 2023). Notably, enhancing Nrf2 activity has been shown to provide neuroprotection by counteracting oxidative stress in AD (Fao et al. 2019; Villavicencio Tejo and Quintanilla 2021).

*Ganoderma lucidum*, an edible medicinal fungus, has long been used in traditional Chinese medicine for its health-promoting and longevity-enhancing properties. Its spores contain bioactive compounds, including polysaccharide and triterpene, which exhibit anti-inflammatory, antitumor, antioxidant, and immunomodulatory effects (Xu and Li 2019; Seweryn et al. 2021). Previous studies have demonstrated that *Ganoderma lucidum* polysaccharides (GLPS) attenuates pathological changes associated with spinal cord injury and ischemia/reperfusion injury and promotes neural progenitor proliferation in the AD mouse models (Zhao et al. 2004; Gokce et al. 2015; Huang et al. 2017). However, the roles of GLPS in modulating oxidative stress and mitochondrial dynamics in AD pathogenesis remain unclear. In this study, the 5  $\times$  FAD transgenic mouse model, a well-established AD model, was used to evaluate the therapeutic effects of GLPS and its underlying mechanisms. The results demonstrated that GLPS treatment improved cognitive function and reduced A $\beta$  and p-tau levels in 5  $\times$  FAD mice. Furthermore, GLPS administration alleviated oxidant stress by decreasing MDA levels and increasing SOD activity. It also mitigated mitochondrial fragmentation by modulating the expression of fission and fusion proteins. Importantly, GLPS enhanced the antioxidant capacity by activating the Nrf2/antioxidant enzymes pathway. These findings suggest that GLPS exerts neuroprotective effects in AD by reducing oxidative stress and regulating mitochondrial dynamics through Nrf2 pathway activation.

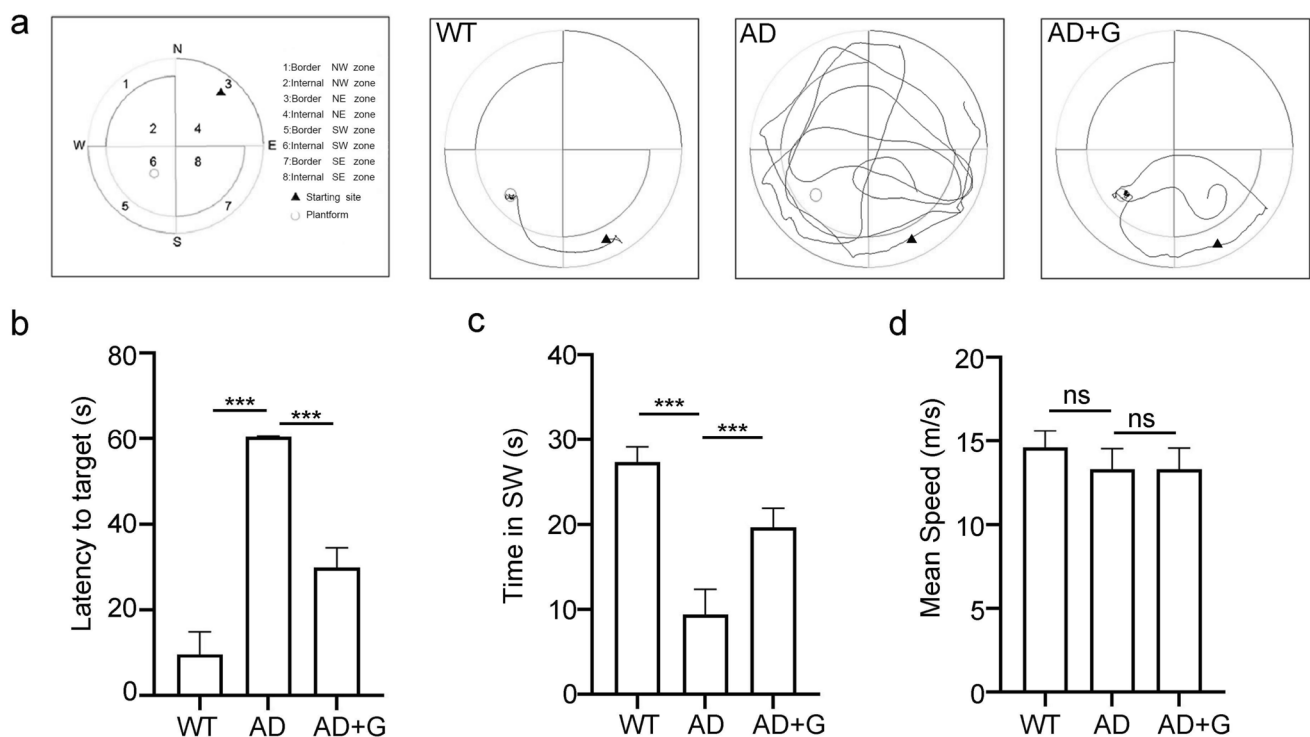
## Materials and methods

### Animals

5  $\times$  FAD transgenic mice were provided by the Institute of Brain Science of Shanxi Datong University. C57BL/6 mice (age: 6–8 months) were sourced from Beijing HFK Bioscience Co., Ltd in Beijing, China. All mice were kept under standardized conditions with 25  $\pm$  2°C temperature, 50  $\pm$  10% humidity, and typical 12-h light/dark cycles. All procedures involving the mice performed in this study were conducted in line with the principles of the Ethics Committee of Shanxi Datong University in Datong, China (No. 2021021).

### Drugs and drug administration

GLPS (license No. H20003510) was provided by Shanxi Guorun Pharmaceutical Co., Ltd in Datong, China. Polysaccharides was extracted from the spore power of *Ganoderma lucidum*. Each ampoule contains 4.5 mg of GLPS in 2 ml of the solution.



**Fig. 1** Morris water-maze (MWM) test was conducted to detect the spatial memory and learning capabilities of the indicated mice. C57BL/6 (age: 6–8 months old) or 5 × FAD mice were administered with normal saline or GLPS for 2 months. **(a)** The schematic diagram of MWM and the representative swimming tracks of the mice in each group during

the training days; **(b)** latency to target of the indicated group of mice in the experimental test; **(c)** time spent by the indicated group of mice in the SW quadrant in the experimental test; **(d)** the mean speed of the indicated group of mice.  $n = 8$  mice per group. Data expressed as the mean  $\pm$  SD. \*\*\*  $p < 0.001$ , ns: no significance

The experimental mice were selected and randomly assigned to three groups of 8 each: C57BL/6 mice were included in the wild type (WT) group, and 5 × FAD transgenic mice were included in the AD and GLPS groups. The dosage and treatment time of GLPS were conducted as per previous studies, albeit with some modifications (Zhu et al. 2007; Huang et al. 2017). For the GLPS group, each mouse was treated with 220  $\mu$ L GLPS (25 mg/kg to 20 g mice) via intraperitoneal injection (i.p.), once a day for 2 months, whereas for WT and AD groups, each mouse was treated with 220  $\mu$ L of normal saline (0.9% sodium chloride) once a day for 2 months. Following drug administration, behavioral tests were performed for 9 days, after which the mice were anesthetized and perfused with cold PBS.

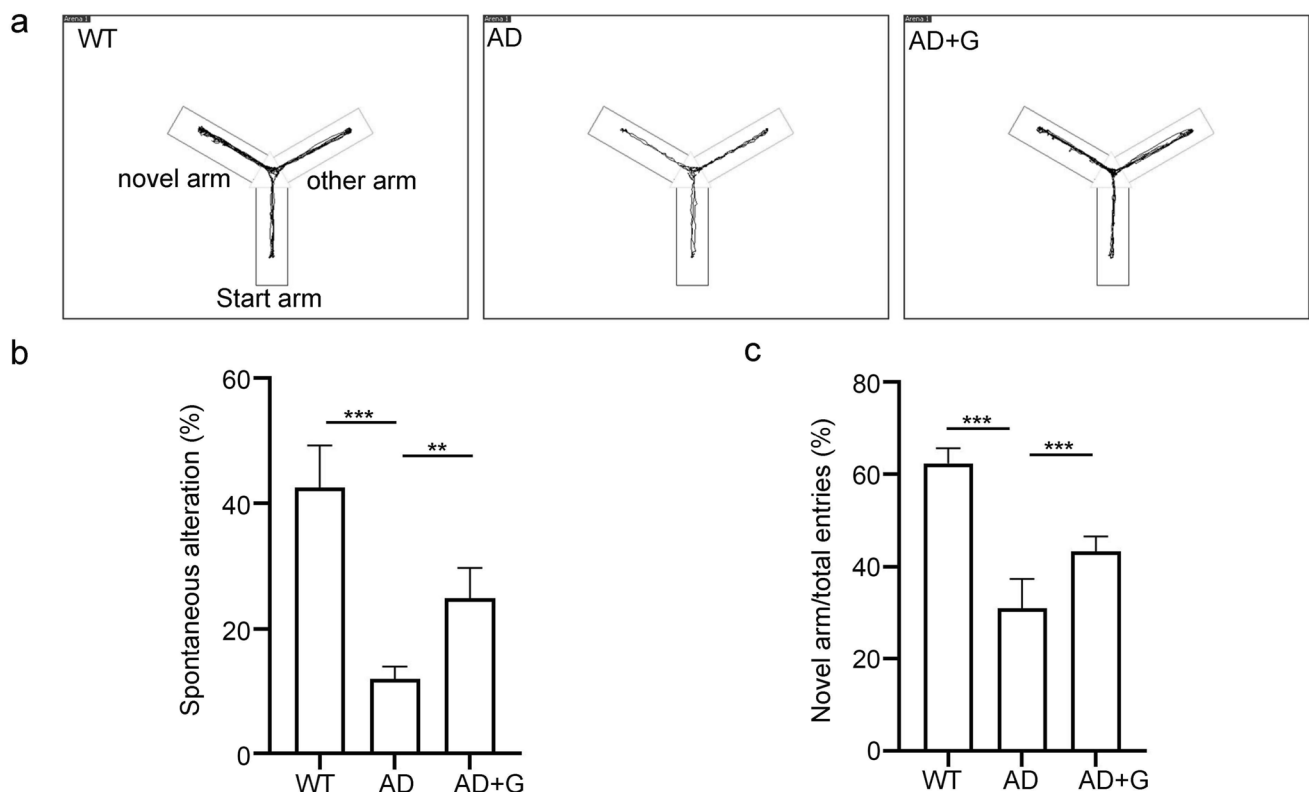
### Morris water-maze (MWM) test

The procedure of MWM is in accordance with a previous protocol, albeit with some modifications (Guo et al. 2020). MWM is a circular pool of diameter 90 cm, and the middle of the SW quadrant has a 5 × 5-cm platform located 2-cm below the water surface. The temperature of opaque water in the pool is maintained at 21–23°C. The mice were trained for 7 consecutive days in the morning. During the training

days, the SE quadrant was set as the start location for the mice. If the mice could not reach the platform within 60 s, they were navigated to the goal and stayed there for 60 s. After 24 h of the last training test, the formal test was conducted and the platform was removed. A water entry point was changed to the NE-quadrant border. The mice were allowed 60 s to swim freely. SMART 3.0 software (Panlab, Barcelona, Spain) was used to track and analyze the activity trajectory of mice and the related parameters.

### Y-maze test

Y-maze tests were administered according to a previously reported method to record the short-term memory of the mice (Guo et al. 2020). Briefly, the Y-maze device includes a start arm, a novel arm, and the other arm. During the training time, the novel arm was blocked, allowing the mice to start at the end of the start arm and then explore the other two arms of the maze for 10 min. After 1 h, the formal test was performed and the barrier at the novel arm was removed. The mice were started at the end of the start arm and allowed to move freely in all three arms for 5 min. Between each trial, the Y-maze was cleaned with 75% ethanol and the new sawdust was replaced to eliminate the taste



**Fig. 2** Y-maze test was performed to test the short-term memory function of the indicated mice. (a) The representative diagrams of mice in each group; (b) the spontaneous alteration of each group of mice in the

Y-maze test; (c) the percentage of choice in the novel arm over all arms of the mice in each group in the Y-maze test.  $n = 8$  mice per group. Data expressed as the mean  $\pm$  SD. \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

left by the previous mouse. The activity track and related parameters of the mice were recorded and analyzed by SMART 3.0 software.

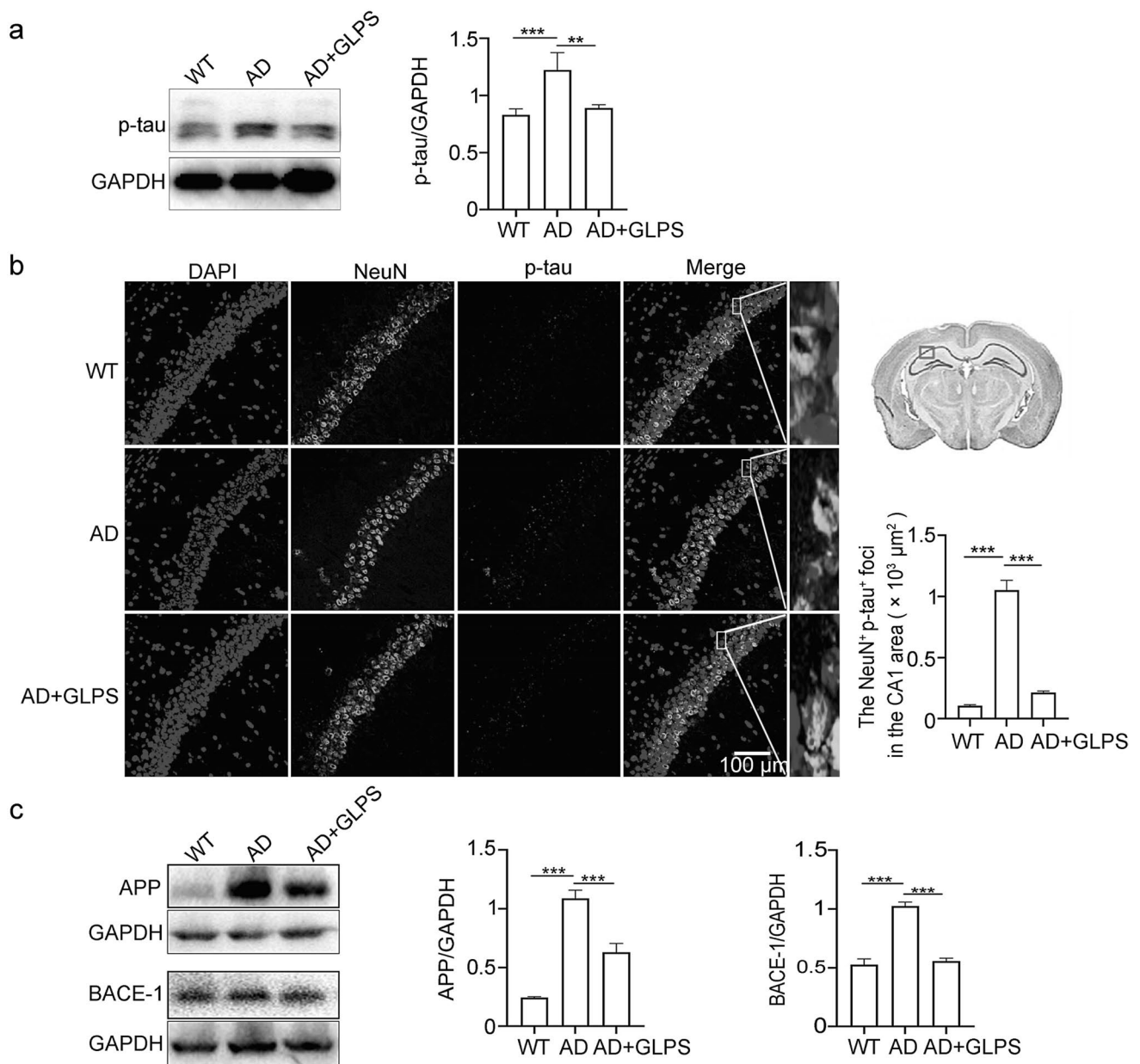
### Immunohistochemistry

The mice ( $n = 4$  per group) were perfused with pre-cold normal saline, followed by 4% paraformaldehyde. The tissue blocks were prepared in liquid nitrogen, and 10- $\mu$ m coronal slices of the brains were sectioned using a cryostat microtome (Leica). For immunohistochemistry, the sections were blocked with 1% bovine serum albumin and 0.3% Triton X-100 in phosphate-buffered saline (PBS) for 2 h at room temperature (RT), followed by incubation with the indicated primary antibodies at 4°C overnight. The primary antibodies used are listed below: the antibodies against p-tau (Ser404) (cat#: 20194S), APP (cat#: 29765S), BACE-1 (cat#: 5606S), A $\beta$  (cat#: 14974 T), Mfn2 (cat#: 9482S), OPA1 (cat#: 80471S), p-Drp1 (Ser616) (cat#: 3455S), Fis1 (cat#: 32525S), SOD2 (cat#: 13141S), NeuN (cat#: 24307S and 94403S) were purchased from Cell Signaling Technology, and the antibodies against Mfn1 (cat#: ab126575), Nrf2 (cat#: ab31163), and NQO1 (cat#: ab80588) were sourced from Abcam, and the antibody against HO-1 (cat#:

ER180273) was sourced from HUABIO. On the second day, the sections were washed thrice with PBS for 5 min and then probed with Alexa Fluor 488 or 594-conjugated secondary antibodies (Abcam, cat#: ab150116 and ab150080) for 2 h at the RT. After rinsing the sections thrice with PBS for 5 min, a mounting medium with antifade and DAPI reagents (Solarbio) was used to mount the slice onto the coverslip. The sections were then photographed with a confocal microscope (CLSM, Olympus, Tokyo, Japan). Image-Pro Plus software was used for the image analysis.

### Western blotting

The mice ( $n = 4$ /group) were anesthetized and perfused with pre-cold normal saline, after which the brains were isolated on ice and immediately frozen at  $-80^{\circ}\text{C}$  until use. For the protein extraction, the brains of mice were removed from the  $-80^{\circ}\text{C}$  storage, sliced into pieces, and lysed with RIPA buffer containing protease inhibitors (i.e., 1 mM PMSF [Roche, cat#: PMSFRO], 1  $\mu\text{g/mL}$  pepstatin [Sigma-Aldrich, cat#: P5318], 2  $\mu\text{g/mL}$  leupeptin [Sigma-Aldrich, cat#: L5793], and 2  $\mu\text{g/mL}$  aprotinin [Roche, cat#: ROAPRO]), and then the samples were sonicated in ice cold water bath (3 s on, 9 s off, 20 cycles). The proteins were collected after



**Fig. 3** GLPS reduced the levels of p-tau, APP, BACE-1, and A $\beta$  in 5  $\times$  FAD mice. **(a)** The tau expressions in the brain tissues of mice in the three groups were detected by WB, and the quantitative results of p-tau were normalized to the GAPDH levels; **(b)** the p-tau expressions were determined by immunohistochemistry staining, and the pixel areas of NeuN<sup>+</sup> p-tau<sup>+</sup> in the hippocampus area CA1 of the mice in the three groups were quantified. DAPI was performed for nucleus staining; **(c)** the APP and BACE-1 levels in the brain tissues of mice in the three groups were detected by WB, and the corresponding quantitative results of APP and BACE-1 were normalized to the GAPDH levels;

**(d)** the A $\beta_{1-42}$  deposition were determined by immunohistochemistry staining, and the pixel areas of A $\beta_{1-42}$  in the hippocampus area CA1 of the mice in the three groups were quantified. DAPI was performed for nucleus staining; The APP **(e)** and BACE-1 **(f)** expressions were determined by immunohistochemistry staining, and the pixel areas of NeuN<sup>+</sup> APP<sup>+</sup> **(e)** and NeuN<sup>+</sup> BACE-1<sup>+</sup> **(f)** in the hippocampus area CA1 of the mice in the three groups were quantified. DAPI was performed for nucleus staining.  $n = 3-4$  mice per groups. Data expressed as the mean  $\pm$  SD. \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

centrifugation at 4°C, 15000 rpm for 30 min. The resulting extract of proteins (50  $\mu$ g) was separated by sodium dodecyl sulfate–polyacrylamide gels and transferred onto polyvinylidene fluoride membranes, which were blocked by 5% non-fat milk at the RT for 1 h and then immunoblotted

by primary antibodies overnight at 4°C. The used primary antibodies are listed below: the antibodies against p-tau (Ser404) (cat#: 20194S), APP (cat#: 29765S), BACE-1 (cat#: 5606S), Mfn2 (cat#: 9482S), OPA1 (cat#: 80471S), p-Drp1 (Ser616) (cat#: 3455S), Fis1 (cat#: 32525S), and

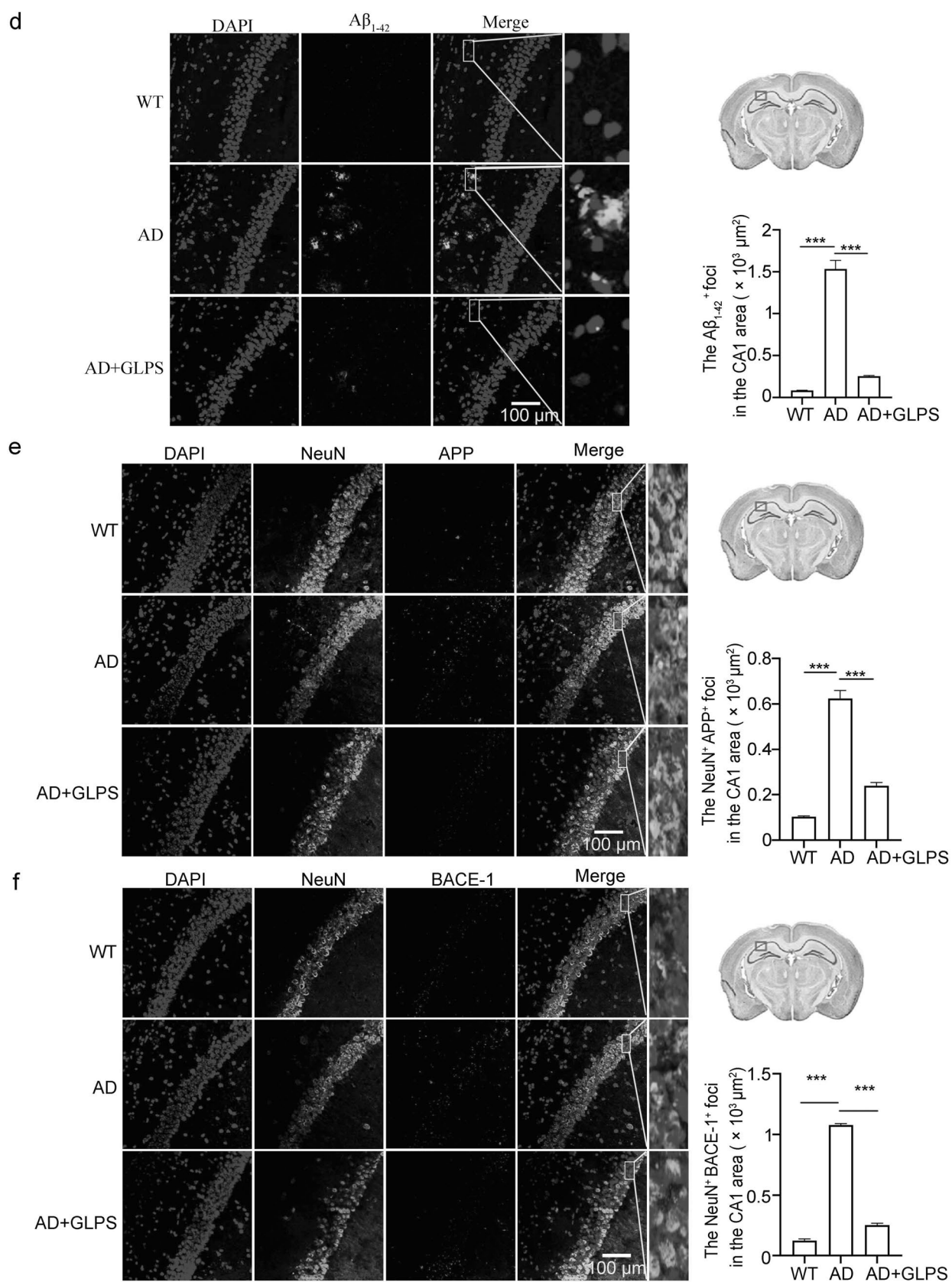
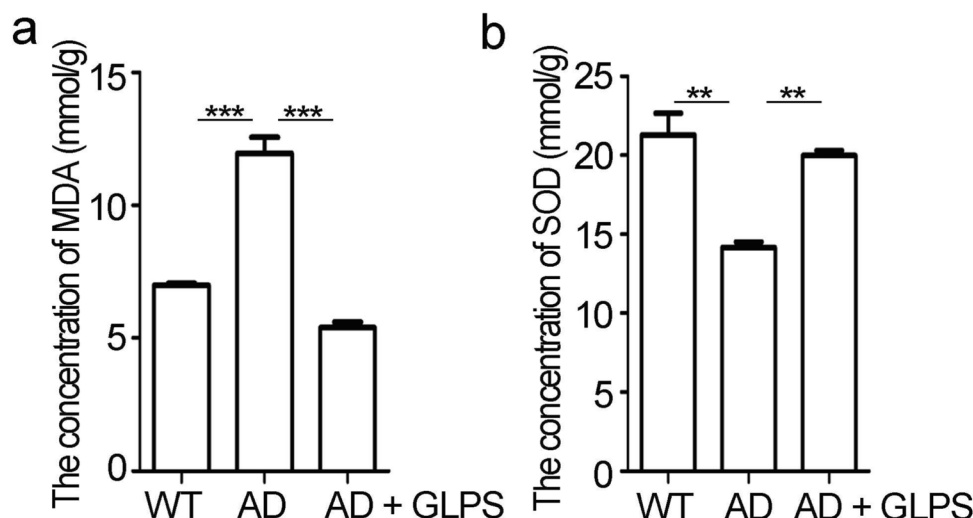


Fig. 3 (continued)

**Fig. 4** GLPS attenuated the oxidative stress in 5 × FAD mice. **(a)** The brain MDA levels of the mice in the three groups are shown; **(b)** the brain SOD levels for each group are shown.  $n = 3$  independent experiments. Data expressed as the mean  $\pm$  SEM. \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$



SOD2 (cat#: 13141S) were purchased from Cell Signaling Technology, the antibodies against Mfn1 (cat#: ab126575), Nrf2 (cat#: ab31163), and NQO1 (cat#: ab80588) were sourced from Abcam, and the antibody against HO-1 (cat#: ER180273) was obtained from HUABIO. The membranes were rinsed thrice with TBST buffer (20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 0.1% Tween- 20) for 8–10 min each time and then incubated with HRP-conjugated corresponding secondary antibodies for 1 h at the RT. After rinsing with the TBST buffer, the protein bands were observed using ECL kits (Bio-Rad). The images were analyzed by Image Lab Software.

### Oxidative stress analysis

For MDA and SOD detections, proteins extracted from mice brain tissues were prepared as per the abovementioned instructions. The protein concentration in the three experimental groups was determined through A280 measurement using the NanoDrop™ software (ThermoFisher), and the samples were diluted to the same concentrations. Then, the MDA and SOD levels were determined by using the MDA kit (Beyotime Biotechnology, cat#: S0109) and the SOD kit (Beyotime Biotechnology, cat#: S0131S).

### Statistical analysis

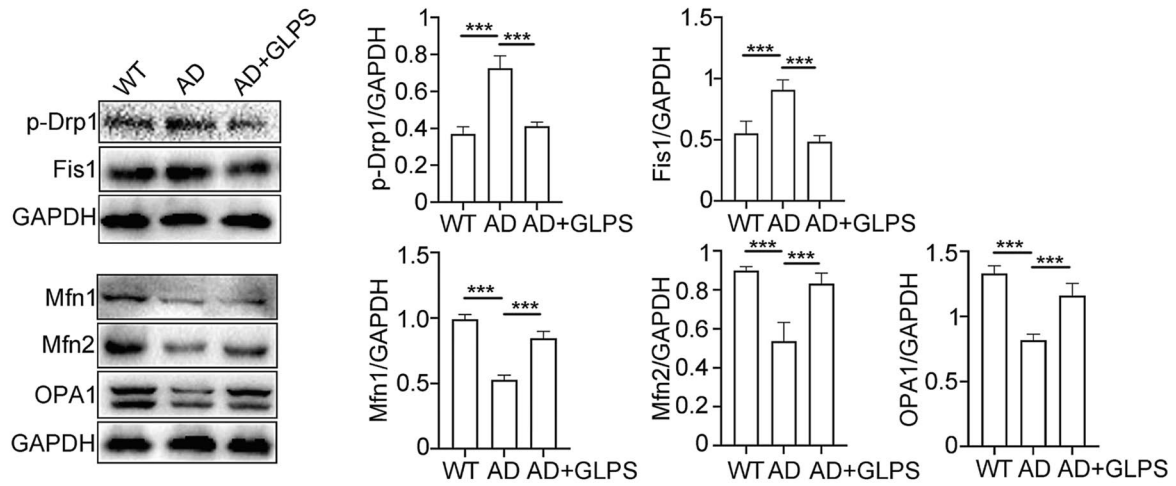
The data analysis was conducted on GraphPad Prism 5.0 software (Graphpad Software, San Diego, CA). The data were expressed as the mean  $\pm$  SD. The statistically significant differences between the different experimental groups were assessed by one-way analysis of variance (ANOVA), followed by Tukey's test for multiple comparisons.  $p < 0.05$  was considered to indicate statistical significance.

## Results

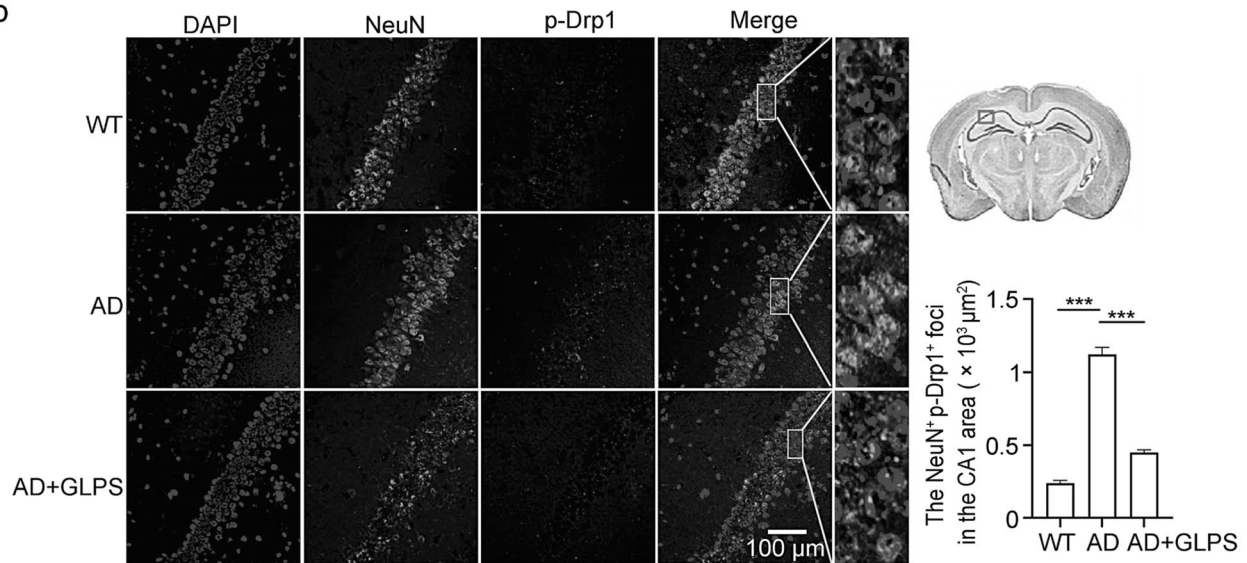
### GLPS ameliorated cognitive impairment in 5 × FAD mice

To assess whether GLPS improves cognitive function in 5 × FAD mice, WT C57BL/6 mice and 5 × FAD mice were administered either normal saline or GLPS (220  $\mu$ L, 2.25 mg/ml) via i.p. for two months. Throughout the treatment period, no significant differences were observed among the groups in diet, body weight, skin color, or fecal appearance. Following treatment, cognitive function was evaluated using the MWM and Y-maze tests. The schematic representation of the MWM and representative swimming traces during the training phase are shown in Fig. 1a. Mice in the AD group exhibited significantly longer latency to reach the target compared to WT mice ( $p < 0.001$ , Fig. 1b), and they spent significantly less time in the southwest (SW) quadrant ( $p < 0.001$ , Fig. 1c). However, GLPS-treated AD mice showed a significantly shorter latency and increased time spent in the SW quadrant compared to untreated AD mice ( $p < 0.001$ ). Motor abilities did not differ significantly among the three groups (Fig. 1d). The schematic representation of the Y-maze test and representative motion traces are shown in Fig. 2a. AD mice exhibited significantly lower spontaneous alternation rates and novel arm choice proportions compared to WT mice. However, GLPS-treated AD mice showed a significant improvement in both parameters ( $p < 0.01$ ,  $p < 0.001$ , respectively) (Figs. 2b–c). These findings indicate that AD mice exhibit impaired spatial learning and memory, which were partially restored by GLPS treatment.

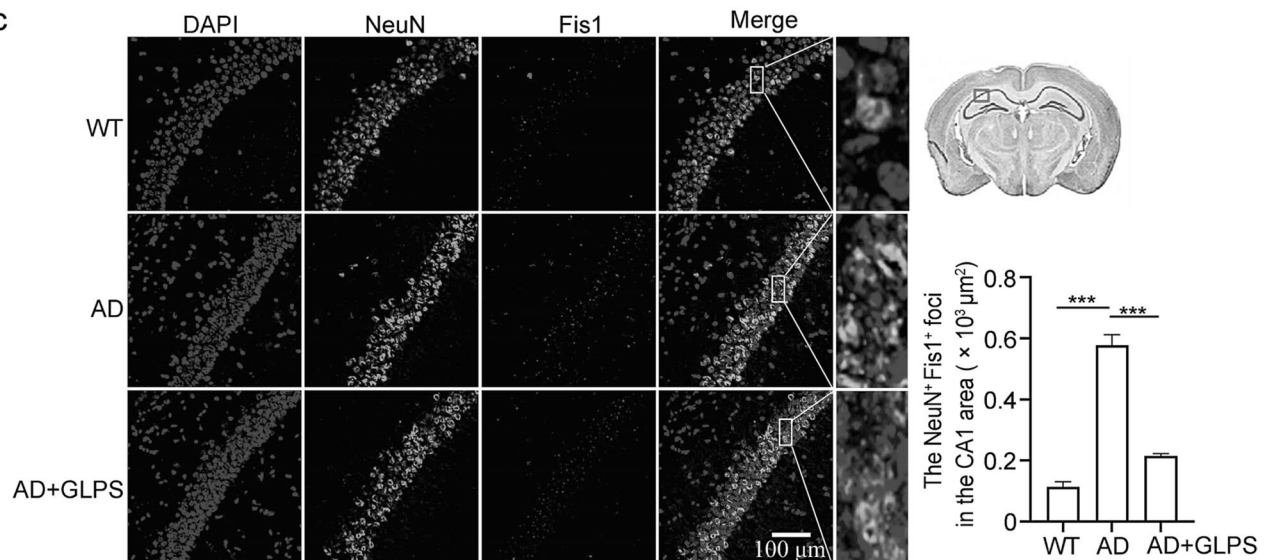
a



b



c



**Fig. 5** GLPS prevented mitochondrial fragmentation and promoted mitochondrial fusion by mediating the expressions of mitochondrial dynamics-associated proteins in 5 × FAD mice. (a) WB analysis of the levels of p-Drp1, Fis1, Mfn1, Mfn2, and OPA1 in each group are shown, and the GAPDH level served as the loading control; the p-Drp1 (b), Fis1 (c), Mfn1 (d), Mfn2 (e), and OPA1 (f) expressions were determined by immunohistochemistry staining, and the pixel areas of p-Drp1<sup>+</sup> (b), NeuN<sup>+</sup>Fis1<sup>+</sup> (c), NeuN<sup>+</sup>Mfn1<sup>+</sup> (d), NeuN<sup>+</sup>Mfn2<sup>+</sup> (e), and NeuN<sup>+</sup>OPA1<sup>+</sup> (f) in the hippocampus area CA1 of the mice in the three groups were quantified. DAPI was performed for nucleus staining. *n* = 3–4 mice per groups. Data expressed as the mean ± SD. \*\*\* *p* < 0.001

### GLPS decreased tau phosphorylation and Aβ deposition in the brain of 5 × FAD mice

A key pathological hallmark of AD is the accumulation of extracellular Aβ plaques and intracellular p-tau-induced neurofibrillary tangles, leading to synaptic and neuronal loss. Aβ is generated from amyloid precursor protein (APP) through enzymatic cleavage, with β-secretase 1 (BACE-1) as the initial cleavage enzyme (Koelsch 2017). To determine whether GLPS affects these pathological features, we assessed the expression levels of p-tau, Aβ, APP, and BACE-1 in the brain. Western blot (WB) analysis revealed significantly elevated p-tau levels in AD mice compared to WT mice (*p* < 0.001), which were significantly reduced following GLPS treatment (*p* < 0.01, Fig. 3a). These findings were consistent with immunohistochemistry results (*p* < 0.001, Fig. 3b). Similarly, APP and BACE-1 levels were significantly increased in AD mice (*p* < 0.001) but significantly decreased following GLPS treatment (*p* < 0.001) (Fig. 3c). Immunofluorescence analysis of the hippocampal CA1 region further confirmed increased Aβ, APP, and BACE-1 levels in AD mice compared to WT mice (*p* < 0.001). GLPS treatment significantly reduced the accumulation of these proteins in the CA1 region (*p* < 0.001) (Figs. 3d–f). These findings suggest that GLPS reduces p-tau and Aβ expression in the brains of 5 × FAD mice, potentially through the downregulation of APP and its processing by BACE-1.

### GLPS attenuated oxidative stress in 5 × FAD mice

Oxidative stress plays a central role in AD pathogenesis, and extracts from *G. lucidum* exhibit potent antioxidant, antitumor, and immunomodulatory properties (Shi et al. 2013). MDA, a major product of lipid peroxidation, and SOD, a key antioxidant enzyme, are commonly used as markers of oxidative stress. To evaluate the effects of GLPS on oxidative stress in AD, MDA and SOD levels were measured across experimental groups. MDA concentrations were significantly elevated in the brain tissues of AD mice compared to WT mice (*p* < 0.001), but GLPS treatment for two months significantly reduced these levels (*p* < 0.001, Fig. 4a). Conversely, SOD levels were significantly lower in AD mice

(*p* < 0.01) but were significantly elevated following GLPS administration (*p* < 0.01) (Fig. 4b). These findings suggest that GLPS mitigates oxidative stress in 5 × FAD mice by reducing lipid peroxidation and enhancing antioxidant enzyme activity.

### GLPS suppressed mitochondrial fragmentation and promoted mitochondrial fusion in 5 × FAD mice

Mitochondria are a major source of ROS in cells. Mitochondrial dysfunction, including disrupted dynamics, can impair metabolic homeostasis and lead to excessive ROS production, thereby inducing oxidative stress (Bhatti et al. 2017). Conversely, elevated ROS levels can trigger mitochondrial fission (Brillo et al. 2021). Both abnormal mitochondrial dynamics and oxidative stress contribute to AD pathogenesis. Thus, we examined the effects of GLPS on mitochondrial dynamics in 5 × FAD mice. WB analysis revealed that fission-related proteins, p-Drp1 and Fis1, were significantly upregulated in AD mice, whereas fusion-related proteins, Mfn1, Mfn2, and OPA1, were significantly downregulated. GLPS treatment reversed these changes (*p* < 0.001, Fig. 5a). Immunofluorescence analysis further confirmed that p-Drp1 and Fis1 levels were significantly elevated in the hippocampal CA1 region of AD mice, while Mfn1, Mfn2, and OPA1 levels were significantly reduced. These alterations were reversed upon GLPS administration (*p* < 0.001) (Figs. 5b–f). These findings indicate that GLPS treatment effectively prevents mitochondrial fragmentation and enhances mitochondrial fusion in 5 × FAD mice, potentially mitigating mitochondrial dysfunction in AD.

### GLPS increased Nrf2/antioxidant enzyme expression in 5 × FAD mice

Accumulative evidence demonstrated that Nrf2-mediated antioxidant enzyme expression exerted neuroprotective effects against cellular oxidative stress and mitochondrial impairment during AD pathogenesis (Villavicencio Tejo and Quintanilla 2021). The underlying mechanism responsible for GLPS treatment improved cognitive deficits in 5 × FAD mice was subsequently explored. As illustrated in Fig. 6a, statistical decreases were detected in the levels of Nrf2, HO-1, NQO1, and SOD2 in the brain of the AD group compared to those in the WT group using WB assay (*p* < 0.01, *p* < 0.001, respectively), and GLPS treatment statistically elevated the expression of the corresponding proteins in brain tissues of AD mice (*p* < 0.05, *p* < 0.01, *p* < 0.001, respectively). Consistent with WB analysis, immunofluorescence staining identified that the Nrf2, HO-1, NQO1, and SOD2 levels in the CA1 region of the hippocampus of the AD group were significantly inhibited compared to the

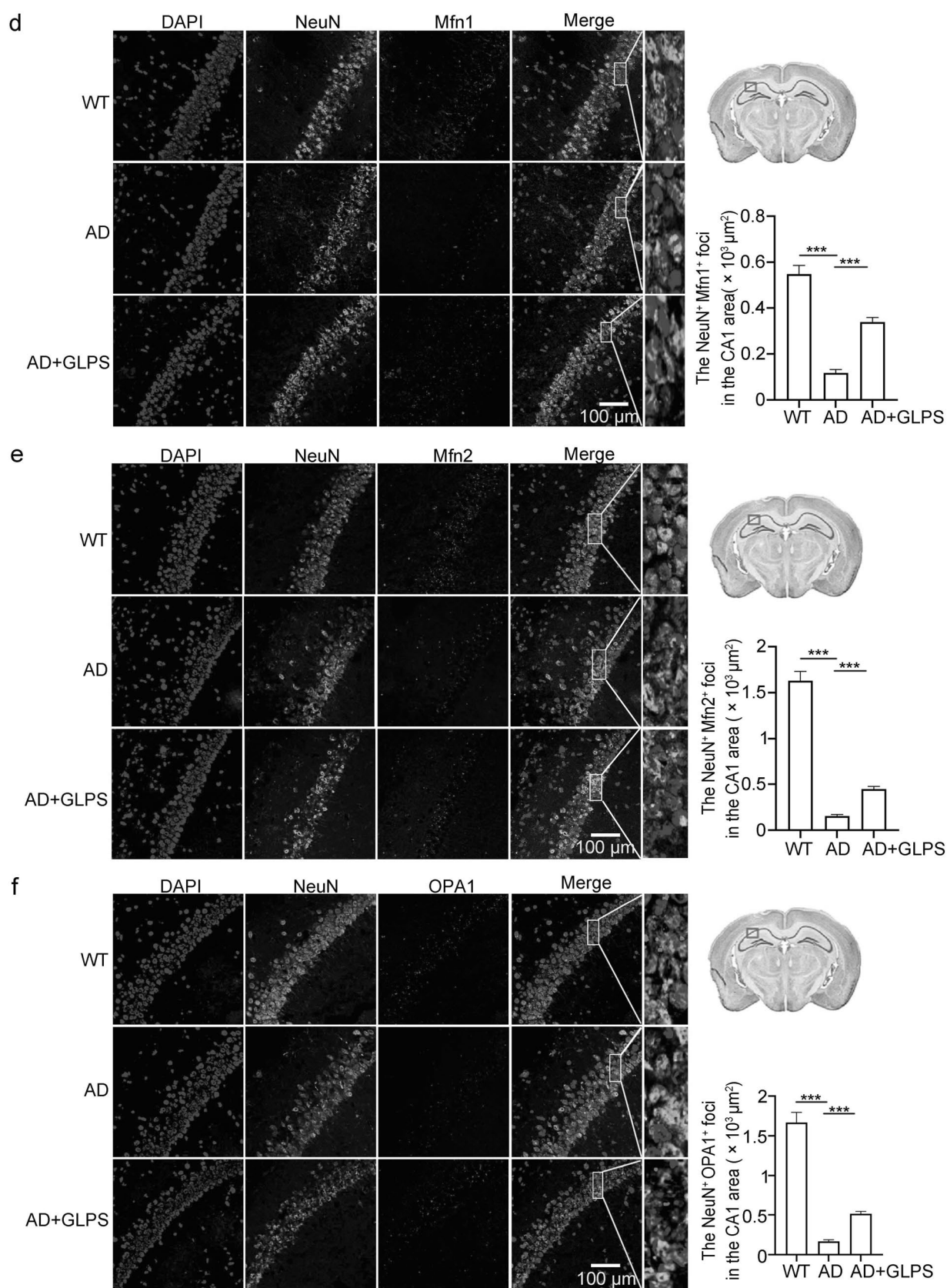


Fig. 5 (continued)

WT group ( $p < 0.001$ ), which were reversed by GLPS treatment ( $p < 0.01$ ,  $p < 0.001$ , respectively) (Figs. 6b–e). These results indicated that GLPS treatment inhibited the activation of the Nrf2/antioxidative axis and played protective roles against neurotoxicity in  $5 \times$  FAD mice.

## Discussion

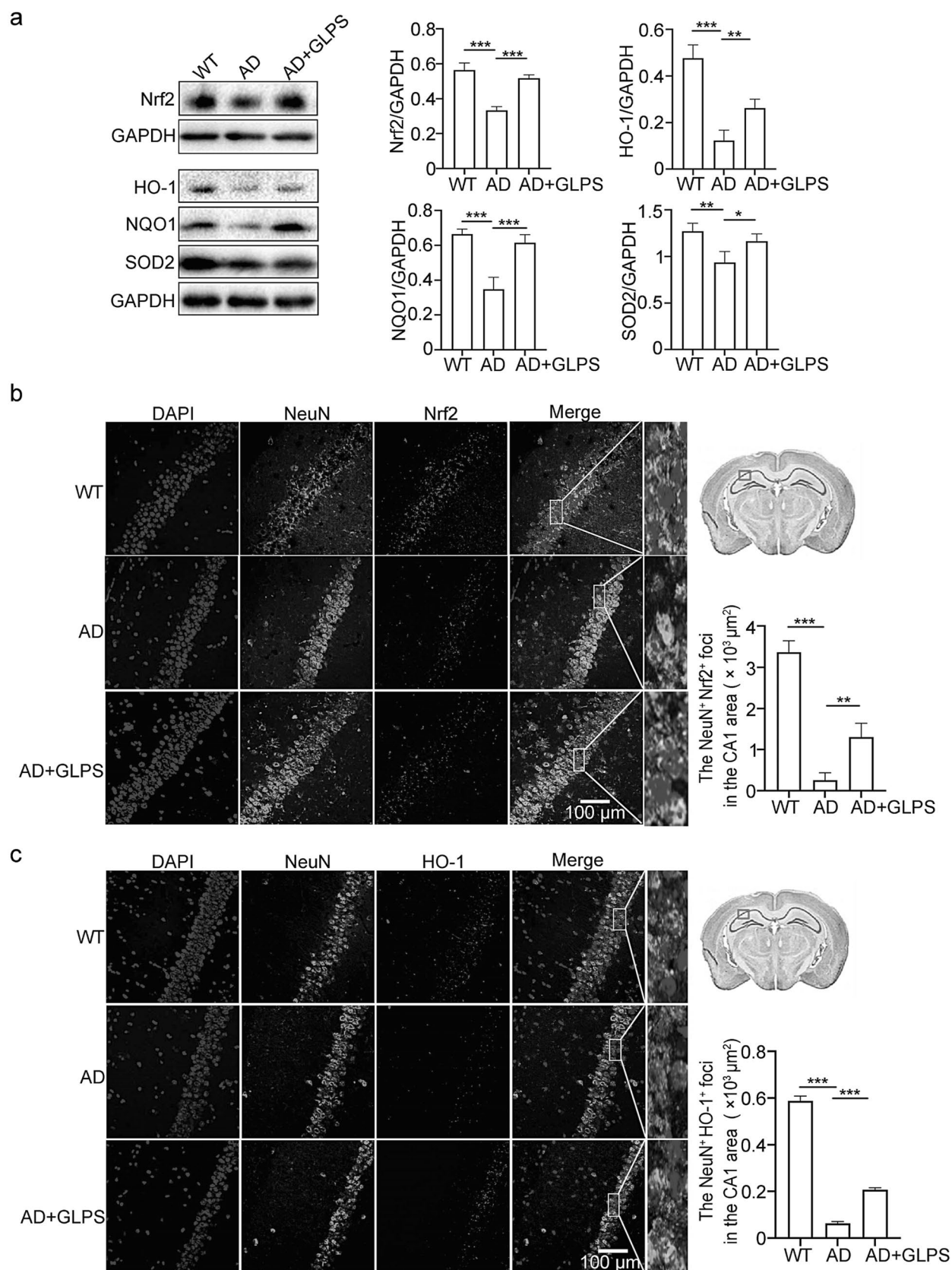
AD is an age-related neurodegenerative disorder characterized by complex pathological processes, including impaired A $\beta$  clearance, oxidative stress, and mitochondrial dysfunction. *Ganoderma lucidum*, commonly known as “Lingzhi”, “Reishi”, or “Youngzhi”, has been used for centuries to promote longevity and overall health. Several studies have demonstrated that the polysaccharides extracted from *Ganoderma lucidum* alleviates the oxidant stress induced by hypoxia/reoxygenation and spinal cord injury (Zhao et al. 2004; Gokce et al. 2015). However, the roles of GLPS in AD pathogenesis remain unclear. The  $5 \times$  FAD mouse model, a widely used transgenic AD model, carries five familial AD mutations in *APP695* (Swedish (K670 N, M671L, Florida I716), and London V717I) and *PSEN1* (M146L and L286 V) on a C57BL/6 J background (Oblak et al. 2021). Given that mitochondrial dynamics play a crucial role in energy metabolism and ROS production, we investigated the effects of GLPS on oxidative stress and mitochondrial homeostasis in  $5 \times$  FAD mice.

GLPS used in this study was provided by Shanxi Guorun Pharmaceutical Co., Ltd (Datong, China) (license No. H20003510). Clinically, GLPS has been widely utilized for treating neurosis, polymyositis, dermatomyositis, atrophic myotonia, and progressive muscular dystrophy, and immune-related disorders. According to the drug's instructions, the standard human dosage is 2 mL per day via intramuscular injection for a treatment duration of 1 to 3 months or as prescribed by a physician. For animal research, factors such as drug source, purity, administration route, and experimental objectives influence dosage selection. Shichao Huang et al. reported that 5- to 6-month-old APP/PS1 mice treated with 30 mg/kg GLP via oral gavage for 90 days exhibited improved learning and spatial memory (Huang et al. 2017). Additionally, Zhu et al. (2007) found that the immunosuppressed mice injected intraperitoneally with 2.5 mg/kg GLPS for seven days exhibited immune cell activity, with no evident toxic effects at doses of 25 mg/kg and 250 mg/kg. In this study,  $5 \times$  FAD mice received daily i.p. of 220  $\mu$ L GLPS (equivalent to 25 mg/kg to a 20 g mouse) for two months, and no overt toxicity was observed. However, as we did not conduct specific toxicity assessments, quantitative data on the potential toxic effects of GLPS in  $5 \times$  FAD mice are lacking. Future studies will be necessary to

comprehensively evaluate GLPS toxicity and further elucidate its underlying mechanisms in AD pathology.

Studies have demonstrated that GLPS exerts neuroprotective effects by promoting neural progenitor cell regeneration and reducing A $\beta$ -induced cytotoxicity (Lai et al. 2008; Huang et al. 2017), suggesting its potential role in mitigating AD pathology. In this study, the MWM and Y-maze tests were used to assess the effects of GLPS on learning and memory in  $5 \times$  FAD mice. The results indicated that GLPS treatment significantly improved cognitive function in these mice. To further evaluate its impact on AD-related pathology, we examined key pathological hallmarks of the disease. GLPS treatment significantly reduced p-tau and A $\beta$  levels in the brain tissues of  $5 \times$  FAD mice, findings consistent with the observed improvements in cognitive performance. Since A $\beta$  is generated through the sequential cleavage of APP, initiated by BACE-1 and followed by  $\gamma$ -secretase, we also assessed the levels of APP and BACE-1. GLPS treatment led to a downregulation of APP and BACE-1 expression in  $5 \times$  FAD mice, suggesting that its beneficial effects on A $\beta$  clearance may result from reduced APP levels and decreased BACE-1-mediated cleavage of APP.

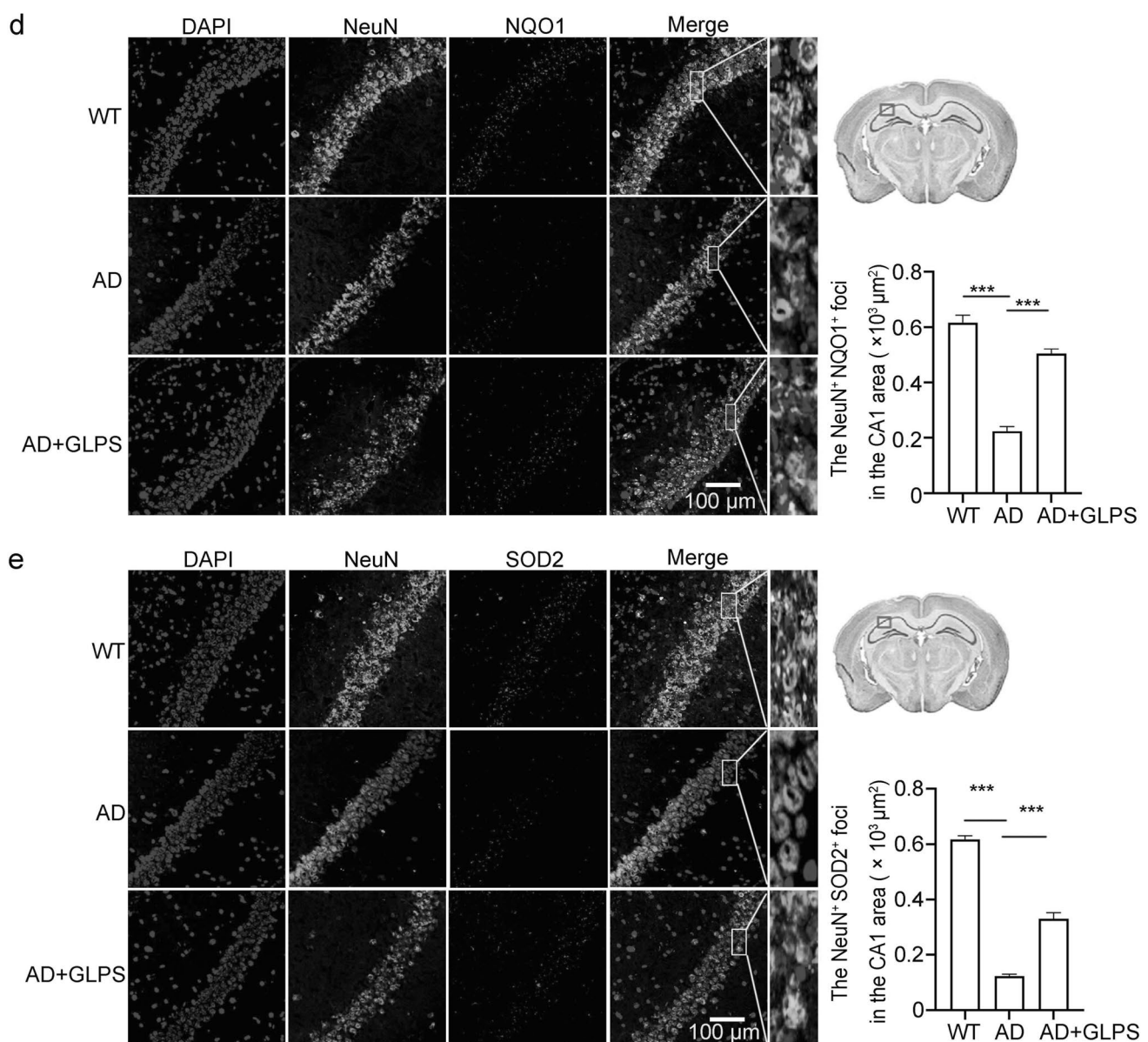
Oxidant stress results from an imbalance between oxidants and antioxidants, leading to excessive ROS production. Due to its high lipid compound and significant energy demands, the brain is particularly susceptible to oxidant damage, making oxidative stress a key contributor to neurodegenerative diseases, including AD. GLPS, a natural polysaccharide, has garnered increasing research attention for its antioxidant properties. The antioxidant activity of polysaccharides is influenced by factors such as monosaccharide composition, side-chain structure, glycosidic bonds, molecular size, and overall structural conformation (Yin et al. 2021). GLPS is heteropolysaccharide composed of glucose, mannose, xylose, galactose and glucuronic acid, with a main chain structure formed by glucose linked through  $\beta$ -glycosidic bonds (Zhao et al. 2023a, b).  $\beta$ -(1  $\rightarrow$  3)-glucan, a major bioactive component of GLPS, has been shown to reduce ROS production (Zhong et al. 2024). Zhao et al. reported that GLPS isolated from fermented *Ganoderma lucidum* spore powder exhibited increased mannose, galactose, and glucuronic acid contents compared to that from unfermented spore powder, leading to enhanced antioxidant activity (Zhao et al. 2023a, b). This suggests that mannose, galactose, and glucuronic acid contribute to the ability of GLPS to mitigate cellular oxidative stress, consistent with studies indicating that polysaccharide antioxidant activity positively correlates with mannose and uronic acid content (Sun et al. 2018; Yuan et al. 2019; Zhong et al. 2024). Additionally, the spatial conformation of polysaccharides is a critical determinant of their antioxidant activity (Yin et al. 2021). However, due to current technological limitations,



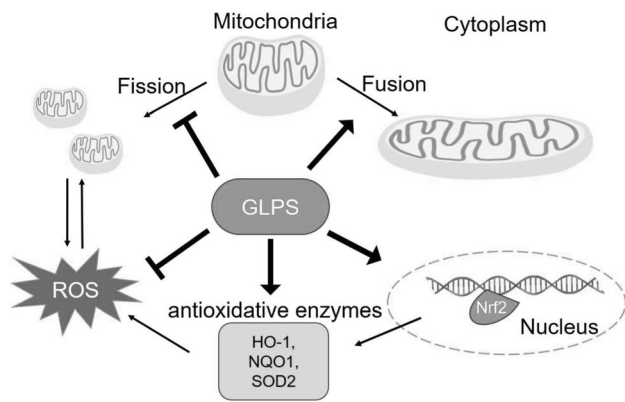
**Fig. 6** GLPS increased the anti-oxidative effects in 5×FAD mice by upregulation of the levels of Nrf2 and its downstream HO-1, NQO1, and SOD2. **(a)** The expressions of Nrf2, HO-1, NQO1, and SOD2 isolated from brain tissues in each group were determined by WB, and the quantitative results of these proteins were normalized to the GAPDH levels; immunohistochemical staining was performed to analyze the expressions of Nrf2 **(b)**, HO-1 **(c)**, NQO1 **(d)**, and SOD2 **(e)** in NeuN<sup>+</sup> cells, and the pixel areas of NeuN<sup>+</sup>Nrf2 **(b)**, NeuN<sup>+</sup>HO-1 **(c)**, NeuN<sup>+</sup>NQO1 **(d)**, and NeuN<sup>+</sup>SOD2 **(e)** in the hippocampus area CA1 of the mice in the three groups were quantified. DAPI was performed for nucleus staining. *n* = 3–4 mice per groups. Data expressed as the mean ± SD. \*\*\* *p* < 0.001

the relationship between GLPS spatial structure and its antioxidant properties remains incompletely understood (Zhong et al. 2024). Nevertheless, GLPS has demonstrated strong antioxidant capacity and neuroprotective effects in

neurodegenerative diseases. For instance, GLPS pretreatment significantly reduced MDA levels and increased SOD activity in hypoxia/reoxygenation-treated rat cortical neurons (Zhao et al. 2004). Similarly, GLPS provided notable neuroprotection against hydrogen peroxide-induced oxidative stress in cerebellar granule cells (Sun et al. 2017). In the present study, we assessed the antioxidant capacity of GLPS in the context of AD. The results showed that MDA levels were significantly elevated, while SOD expression was significantly reduced in AD mice compared to WT mice. However, GLPS administration effectively lowered MDA concentrations and increased SOD expression in AD mice, suggesting that its neuroprotective effects in AD are closely associated with its antioxidant properties.



**Fig. 6** (continued)



**Fig. 7** Schematic depiction of the roles of GLPS in the regulation of mitochondrial dynamics, ROS production, and Nrf2/antioxidant axis in  $5 \times$  FAD mice

Mitochondria serve as the central hub for cellular energy production, and excessive ROS generation is primarily linked to mitochondrial dysfunction. Aberrant mitochondrial dynamics are a hallmark of mitochondrial dysfunction, often accompanied by structural disruption and further ROS overproduction. Previous studies have demonstrated that oxidative stress-induced ROS can promote Drp1 phosphorylation, leading to mitochondrial fragmentation. Excessive mitochondrial fragmentation has been implicated in neuronal death during AD progression, and inhibition this process has been suggested as a potential therapeutic strategy for AD (Wang et al. 2020; Zhang et al. 2023). To further investigate the roles of GLPS in regulating mitochondrial dynamics in  $5 \times$  FAD mice, we examined the expression levels of key mitochondrial fission and fusion proteins. The results showed that GLPS administration reduced the levels of p-Drp1 and Fis1 while upregulating Mfn1, Mfn2, and OPA1 expression in the hippocampal CA1 region and brain tissues of  $5 \times$  FAD mice. These findings suggest that GLPS inhibits mitochondrial fragmentation in AD pathology, potentially contributing to the attenuation of oxidative stress in AD.

Nrf2 is a transcription factor that regulates the expression of antioxidant enzymes to mitigate cellular oxidative stress. Numerous studies have confirmed that activation of the Nrf2/antioxidant enzyme pathway can alleviate oxidative stress and provide therapeutic benefits in AD (Brandes and Gray 2020; Villavicencio Tejo and Quintanilla 2021). For instance, fasudil, a Rho-associated protein kinase inhibitor, significantly improved cognitive function and enhanced Nrf2 signaling in APP/PS1 mice (Wei et al. 2021). Moreover, both genetic and pharmacological induction of Nrf2 has been shown to ameliorate cognitive impairment in AD mouse models (Uruno et al. 2020). Additionally, a previous study reported that Nrf2 expression is downregulated in  $5 \times$  FAD mice (Ullah et al. 2021). To elucidate the molecular mechanisms underlying GLPS-mediated regulation of

oxidative stress and mitochondrial dynamics in  $5 \times$  FAD mice, we examined the expression levels of Nrf2, HO-1, NQO1, and SOD2. AD mice exhibited significantly reduced levels of Nrf2, HO-1, NQO1, and SOD2, which were inhibited following GLPS administration. These results suggest that GLPS attenuates oxidative damage in  $5 \times$  FAD mice by activating the Nrf2/antioxidant enzyme signaling pathway. However, the precise mechanism by which GLPS promotes Nrf2 translocation from the cytoplasm to the nucleus remains unclear and warrants further investigation.

In conclusion, this study demonstrated that GLPS, a naturally derived polysaccharide from *Ganoderma lucidum*, alleviated cognitive impairment in  $5 \times$  FAD mice and reduced p-tau and A $\beta$  levels in their brains. Furthermore, GLPS treatment mitigated oxidative damage and excessive mitochondrial fragmentation, potentially through activation of the Nrf2/antioxidant enzyme pathway (Fig. 7). These findings suggest that GLPS holds promise as a neuroprotective agent for preventing the onset and progression of AD.

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**Author contributions** Xiaoqin Liu and Yanbing Li performed the most of assays, including the experimental design and execution, and results analysis. Xiaoqin Liu completed the original draft writing and Cungen Ma further reviewed and edited the manuscript. Jiwei Wang, Tao Meng, Lijuan Song and Lizhi Yang helped with mice related procedures and confocal photomicrographs experiments. Jiezhong Yu and Cungen Ma contributed to the supervision of the study.

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**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Ethics approval** This study was performed in line with the principles of the Ethics Committee of Shanxi Datong University in Datong, China (No. 2021021).

**Consent to participate** Not applicable.

**Consent to publish** Not applicable.

**Competing interests** The authors declare no competing interests.

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